Robust Beamforming Optimization for Cell-Free ISAC Systems Mathematical Derivation

1 Original Problem

The decision variables of the original problem (P1) are the per-AP beamforming vectors $\mathbf{w}_{m,k} \in \mathbb{C}^{N_t}$, the sensing covariance matrices $\mathbf{Z}_m \in \mathbb{H}_+^{N_t}$, and the AP selection indicators $b_{mp} \in \{0,1\}$:

$$\min_{\substack{\{\mathbf{w}_{m,k}\}_{m\in\mathcal{M},k\in\mathcal{K}},\{\mathbf{Z}_m\}_{m\in\mathcal{M}},\\ \{b_{mp}\}_{m\in\mathcal{M},p\in\mathcal{P}}}} \quad \sum_{m=1}^{M}\Big(\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|^2 + \mathrm{tr}(\mathbf{Z}_m)\Big) \tag{P1-objective}$$

$$\text{s.t.} \quad \mathrm{SINR}_{S,p} := \frac{|\mathbf{g}_p^H \mathbf{Z}\mathbf{g}_p|}{\sigma_s^2} \geq \gamma_S^{\mathrm{PoD}}, \quad \forall p \in \mathcal{P} \tag{P1-C2}$$

$$\mathrm{tr}(\mathbf{J}_p^{\mathrm{data}}) \geq \Gamma_{\mathrm{Track},p}, \quad \forall p \in \mathcal{P} \tag{P1-C3}$$

$$\sum_{m=1}^{M} b_{mp} = N_{\mathrm{req}}, \quad \forall p \in \mathcal{P}^{\mathrm{active}} \tag{P1-C4}$$

$$\sum_{k=1}^{K}\|\mathbf{w}_{m,k}\|^2 + \mathrm{tr}(\mathbf{Z}_m) \leq P_{\max}, \quad \forall m \in \mathcal{M} \tag{P1-C5}$$

$$\mathbf{Z}_m \succeq \mathbf{0}, \quad \forall m \in \mathcal{M} \tag{P1-C6}$$

$$b_{mp} \in \{0,1\}, \quad \forall m \in \mathcal{M}, p \in \mathcal{P} \tag{P1-C7}$$

The communication SINR (P1-C1) and the worst-case SINR are defined as (**Note: the fractional structure and the semi-infinite uncertainty are the core non-convex sources**):

$$\mathrm{SINR}_k^{\mathrm{wc}} := \min_{\|\Delta\mathbf{h}_k\|\leq\epsilon_h\|\hat{\mathbf{h}}_k\|} \frac{|(\hat{\mathbf{h}}_k + \Delta\mathbf{h}_k)^H\mathbf{w}_k|^2}{\sum_{j\neq k}|(\hat{\mathbf{h}}_k + \Delta\mathbf{h}_k)^H\mathbf{w}_j|^2 + \sigma_c^2} \geq \gamma_k, \quad \forall k \tag{P1-C1}$$

where $\mathbf{w}_k \triangleq [\mathbf{w}_{1,k}^T, \mathbf{w}_{2,k}^T, \ldots, \mathbf{w}_{M,k}^T]^T \in \mathbb{C}^{MN_t}$ is the cooperative stacked beamforming vector from all APs to user k (**Note**: the decision variables are the per-AP vectors $\mathbf{w}_{m,k}$, and $\mathbf{w}_k$ is obtained by stacking them).

2 Covariance Lifted Form (P2) — Equivalent Lifted Formulation

To prepare for the semidefinite relaxation, we lift the vector variables in (P1) into covariance matrices, obtaining **(P2)**:

$$
\begin{aligned}
\min_{\substack{\{\mathbf{W}_k\}_{k\in\mathcal{K}},\mathbf{Z}\in\mathbb{H}_+^{MN_t},\\ \{b_{mp}\}_{m\in\mathcal{M},p\in\mathcal{P}}}} \quad & \sum_{k=1}^{K}\mathrm{tr}(\mathbf{W}_k)+\mathrm{tr}(\mathbf{Z}) && \text{(P2)}\\
\text{s.t.} \quad & \text{(P1-C1) reformulated as (P2-C1)} && \text{(P2-C1)}\\
& \mathrm{tr}(\mathbf{g}_p\mathbf{g}_p^H\mathbf{Z}) \geq \gamma_S^{\mathrm{PoD}}\sigma_s^2, \quad \forall p && \text{(P2-C2)}\\
& \mathrm{tr}(\mathbf{F}_p\mathbf{R}_X) \geq \Gamma_{\mathrm{Track},p}, \quad \forall p && \text{(P2-C3)}\\
& \mathrm{tr}(\mathbf{E}_m\mathbf{R}_X) \leq P_{\max}, \quad \forall m && \text{(P2-C4)}\\
& \mathbf{W}_k \succeq \mathbf{0}, \quad \forall k && \text{(P2-C5)}\\
& \mathbf{Z} \succeq \mathbf{0} && \text{(P2-C6)}\\
& \mathrm{rank}(\mathbf{W}_k)=1, \quad \forall k && \text{(P2-C7)}\\
& b_{mp}\in\{0,1\}, \quad \forall m,p && \text{(P2-C8)}
\end{aligned}
$$

where:

- $\mathbf{W}_k = \mathbf{w}_k\mathbf{w}_k^H \in \mathbb{H}_+^{MN_t}$, $\mathbf{w}_k = [\mathbf{w}_{1,k}^T,\ldots,\mathbf{w}_{M,k}^T]^T$
- $\mathbf{R}_X \triangleq \sum_{k=1}^{K}\mathbf{W}_k + \mathbf{Z}$
- $\mathbf{E}_m = \mathrm{diag}(\underbrace{0,\ldots,0}_{(m-1)N_t},\underbrace{1,\ldots,1}_{N_t},\underbrace{0,\ldots,0}_{(M-m)N_t})$ is the AP m selection matrix
- $\mathbf{F}_p = \frac{2}{\sigma_s^2}\mathrm{Re}\{\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p\} \in \mathbb{H}_+^{MN_t}$
- (P2-C1) is the LMI form of (P1-C1) after the S-Procedure; see §4 Step 3

(P1) ⇔ (P2) is a strict equivalence: the variable mapping $\mathbf{w}_k \leftrightarrow \mathbf{W}_k = \mathbf{w}_k\mathbf{w}_k^H$ is a one-to-one correspondence (the rank-one constraint guarantees recoverability), and all constraints are reformulated equivalently via identities such as $\mathrm{tr}(\mathbf{x}\mathbf{x}^H\mathbf{W}) = \mathbf{x}^H\mathbf{W}\mathbf{x}$. **No relaxation is involved**—it is purely a mathematical rewriting.

3 Non-Convexity Analysis

Note that problem (P2) is non-convex. Specifically, the SINR constraint (P2-C1) involves a fractional quadratic form in $\mathbf{W}_k$ with semi-infinite worst-case uncertainty (NC2), making the universal quantifier "$\forall\|\Delta\mathbf{h}_k\| \leq \epsilon_h\|\hat{\mathbf{h}}_k\|$" non-convex; the per-AP power constraint (P2-C4) couples the per-AP beamforming vectors $\mathbf{w}_{m,k}$ with the sensing covariance $\mathbf{Z}_m$ (NC5); the PCRB constraint (P2-C3) involves the trace of an inverse Fisher information matrix that depends nonlinearly on $\mathbf{R}_X$ (NC6); the AP-selection constraint (P2-C8) requires a binary indicator $b_{mp} \in \{0,1\}$ (NC3); and the rank-one constraint (P2-C7) demands a non-convex rank-one manifold (NC4). The PSD constraints (P2-C5)/(P2-C6) and the per-AP power linearization are themselves convex, but the overall problem remains non-convex due to the coupling identified above. In the following section, we apply a six-step convexification chain to handle each non-convexity in turn, ultimately reducing the problem to a standard convex SDP.

4 Step-by-Step Convexification

We apply six transformations to (P2), progressively converting each non-convexity into a convex form. The overall roadmap is: first use SDR to sacrifice rank-one tightness for convexification of the feasible set, then apply the S-Procedure to handle the semi-infinite robust constraints, and finally linearize or retain the PCRB / sensing SINR / power constraints. Each transformation preserves strict equivalence when possible (only Step 1 is a tight relaxation for $K > 2$, and Step 6 is an engineering heuristic), with the equivalence type annotated inline.

We first address the most stubborn non-convexity—the rank-one constraint (P2-C7). Although (P1) $\Leftrightarrow$ (P2) strictly preserves rank-one, the rank-one manifold is itself non-convex. SDR removes this non-convexity by relaxing (P2-C7) to merely $\mathbf{W}_k \succeq \mathbf{0}$:

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k. \tag{S1.1}$$

The feasible set enlarges from the rank-one manifold $\mathcal{F}_{\text{rank-1}}$ to the PSD cone $\mathcal{F}_{\text{SDR}} = \{\mathbf{W} \succeq \mathbf{0}\}$. The SDR provides a lower bound $P^*_{\text{SDR}} \leq P^*_{\text{original}}$ on the original problem. Note that for our per-user SINR + cell-free cooperative + robust problem, **no universal SDR tightness theorem applies**—the $G_k \leq 2$ tightness condition for multi-group multicasting and the $N_t \leq 2$ tightness condition for robust MISO do not transfer. When the SDR solution has $\text{rank}(\mathbf{W}_k^*) > 1$, **Gaussian randomization** is required to recover a feasible rank-1 solution—this is the standard practice in SDR-based beamforming, but **no tight performance-loss bound exists**; engineering practice typically uses $L = 100 \sim 1000$ candidates and selects the best.

After handling rank-one, constraint (P2-C1) still carries a fractional structure and worst-case uncertainty, which must be addressed in two coupled steps: first rewrite the fraction as a quadratic form that the S-Procedure can process, then let the S-Procedure handle the $\Delta\mathbf{h}$ uncertainty. Fixing a worst-case realization $\Delta\mathbf{h}_k$ satisfying $\|\Delta\mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|$, the SINR inequality

$$\frac{\text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) + (\Delta\mathbf{h}\text{ correction term})}{\sum_{j\neq k} \text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) + (\Delta\mathbf{h}\text{ correction term}) + \sigma_c^2} \geq \gamma_k$$

under the assumption that the denominator is strictly positive ($\sigma_c^2 > 0$), can be cross-multiplied to yield the quadratic inequality

$$\text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_k) - \gamma_k \sum_{j\neq k} \text{tr}(\hat{\mathbf{H}}_k \mathbf{W}_j) \geq \gamma_k \sigma_c^2. \tag{S2.1}$$

This step is a strict equivalence, not an approximation—the positivity of the denominator guarantees that multiplication by the denominator is reversible. The resulting quadratic form (S2.1) is exactly the input required by the S-Procedure.

We now restore the worst-case $\Delta\mathbf{h}_k$ to a variable. Defining $\mathbf{A}_k \triangleq \frac{1}{\gamma_k}\mathbf{W}_k - \sum_{j\neq k} \mathbf{W}_j$, constraint (P2-C1) can be rewritten as a quadratic inequality in $\Delta\mathbf{h}_k$:

$$\tilde{f}_1(\Delta\mathbf{h}) = (\hat{\mathbf{h}} + \Delta\mathbf{h})^H \mathbf{A}_k (\hat{\mathbf{h}} + \Delta\mathbf{h}) - \sigma_c^2 \geq 0, \quad \forall \|\Delta\mathbf{h}_k\| \leq \epsilon_h \|\hat{\mathbf{h}}_k\|.$$

The uncertainty set is described by the quadratic constraint $\tilde{f}_2(\Delta\mathbf{h}) = \epsilon_h^2\|\hat{\mathbf{h}}_k\|^2 - \|\Delta\mathbf{h}_k\|^2 \geq 0$. The S-Procedure asserts that, for a single quadratic constraint over a norm ball, the universal quantifier "$\forall\Delta\mathbf{h} \in \mathcal{B}_\epsilon : \tilde{f}_1 \geq 0$" is equivalent to the existence of a multiplier $\mu_k \geq 0$ such that $\tilde{f}_1 - \mu_k\tilde{f}_2 \geq 0$ for all $\Delta\mathbf{h}$, i.e., the quadratic form in $\Delta\mathbf{h}$ is $\succeq 0$—which is captured by the LMI:

$$\exists\mu_k \geq 0 : \begin{bmatrix} \mathbf{A}_k + \mu_k\mathbf{I} & \mathbf{A}_k\hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H\mathbf{A}_k & \hat{\mathbf{h}}_k^H\mathbf{A}_k\hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k\epsilon_h^2\|\hat{\mathbf{h}}_k\|^2 \end{bmatrix} \succeq \mathbf{0}. \tag{S3.1}$$

Sufficiency is immediate ($\tilde{f}_1 - \mu_k\tilde{f}_2 \succeq 0$ directly yields $\tilde{f}_1 \geq \mu_k\tilde{f}_2 \geq 0$ inside the ball); necessity follows from the condition $\mathbf{A}_k + \mu_k\mathbf{I} \succeq \mathbf{0}$ together with the Lagrangian dual. The new slack variable $\mu_k \geq 0$ is the S-Procedure multiplier.

Next we address the sensing-side and tracking-accuracy constraints. The PCRB constraint (P2-C3) involves the trace of the Fisher information matrix—but the FIM itself is **affine** in $\mathbf{R}_X$, because $\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p$ is determined by the target's predicted state in the current time slot and can be treated as a known constant. With $\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p \in \mathbb{C}^{MN_t\times D}$ (D is the target state dimension), $\mathbf{J}_p^{\text{data}} \in \mathbb{C}^{D\times D}$, and its trace is converted into a linear function of $\mathbf{R}_X$ via the cyclic property $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB})$:

$$\begin{aligned} \text{tr}(\mathbf{J}_p^{\text{data}}) &= \frac{2}{\sigma_s^2}\text{Re}\Big\{\text{tr}\big(\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p \cdot \mathbf{R}_X \cdot \nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H\big)\Big\} \\ &= \frac{2}{\sigma_s^2}\text{Re}\Big\{\text{tr}\big(\mathbf{R}_X \cdot \underbrace{\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H \cdot \nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p}_{\in\mathbb{C}^{MN_t\times MN_t}}\big)\Big\}. \end{aligned} \tag{S4.1a}$$

Defining the Hermitian PSD constant matrix

$$\mathbf{F}_p = \frac{2}{\sigma_s^2}\text{Re}\{\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H \cdot \nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p\} \in \mathbb{H}_+^{MN_t}$$

(where $\nabla_{\boldsymbol{\theta}_p}\mathbf{g}_p^H \in \mathbb{C}^{MN_t\times D}$ and $\nabla_{\boldsymbol{\theta}_p}^H\mathbf{g}_p \in \mathbb{C}^{D\times MN_t}$, the outer product is $\mathbb{C}^{MN_t\times MN_t}$; PSD is guaranteed by $\mathbf{x}^H(\mathbf{A}^H\mathbf{A})\mathbf{x} = \|\mathbf{A}\mathbf{x}\|^2 \geq 0$), constraint (P2-C3) becomes equivalent to

$$\text{tr}(\mathbf{F}_p\mathbf{R}_X) \geq \Gamma_{\text{Track},p}, \tag{S4.1b}$$

i.e., (P2-C3) is strictly equivalent to (S4.1b), which is a linear constraint in $\mathbf{W}_k, \mathbf{Z}$.

The sensing SINR constraint (P2-C2) is equally simple in the lifted domain: $\frac{|\mathbf{g}_p^H\mathbf{Z}\mathbf{g}_p|}{\sigma_s^2} \geq \gamma_S^{\text{PoD}}$ can be cross-multiplied directly (σ_s^2 is a constant, and $\mathbf{Z} \succeq \mathbf{0} \Rightarrow \mathbf{g}_p^H\mathbf{Z}\mathbf{g}_p \geq 0$ guarantees a non-negative numerator), yielding

$$\text{tr}(\mathbf{g}_p\mathbf{g}_p^H\mathbf{Z}) \geq \gamma_S^{\text{PoD}}\sigma_s^2, \tag{S4.2}$$

which is linear in $\mathbf{Z}$.

The per-AP power constraint (P2-C4) remains linear in the lifted domain: using $\mathbf{E}_m$ to select the antenna components of AP m ($\mathbf{E}_m \in \mathbb{R}^{MN_t\times MN_t}$ is a diagonal selection matrix), $\sum_k\|\mathbf{w}_{m,k}\|^2 + \text{tr}(\mathbf{Z}_m) = \text{tr}(\mathbf{E}_m\mathbf{R}_X)$, so (P2-C4) is equivalent to

$$\text{tr}(\mathbf{E}_m\mathbf{R}_X) \leq P_{\max}, \tag{S5.1}$$

requiring no further transformation.

Finally, we address the discreteness of the AP-selection constraint (P2-C8). This constraint is NP-hard (a variant of the K-medoid problem), and we adopt a two-step decomposition: the outer step ranks the APs by large-scale fading $\mathrm{PL}(d_{m,p})$ and selects the top-N_{req} APs to fix the service set $\mathcal{M}_p$; the inner step solves a convex SDP on the fixed set. This decomposition **carries no theoretical optimality guarantee**—the outer selection may not be the globally optimal combination (NP-hard)—but engineering practice shows it works well, and the inner SDP on a fixed AP set remains convex and tight.

5 Final Convex SDP Problem (P3) — Convex Relaxation of (P2)

Combining the six transformations from §4, starting from **(P2)** and dropping the rank-one constraint (P2-C7) and applying the heuristic handling of (P2-C8), we obtain the **(P3)** convex SDP:

$$\min_{\{\mathbf{W}_k\},\mathbf{Z},\{\mu_k\}} \quad \sum_{k=1}^{K} \mathrm{tr}(\mathbf{W}_k) + \mathrm{tr}(\mathbf{Z}) \tag{P3}$$

$$\text{s.t.} \quad \begin{bmatrix} \mathbf{A}_k + \mu_k \mathbf{I} & \mathbf{A}_k \hat{\mathbf{h}}_k \\ \hat{\mathbf{h}}_k^H \mathbf{A}_k & \hat{\mathbf{h}}_k^H \mathbf{A}_k \hat{\mathbf{h}}_k - \sigma_c^2 - \mu_k \epsilon_h^2 \end{bmatrix} \succeq \mathbf{0}, \quad \forall k \tag{P3-C1}$$

$$\mathrm{tr}(\mathbf{g}_p \mathbf{g}_p^H \mathbf{Z}) \geq \gamma_S^{\text{PoD}} \sigma_s^2, \quad \forall p \tag{P3-C2}$$

$$\mathrm{tr}(\mathbf{F}_p \mathbf{R}_X) \geq \Gamma_{\text{Track},p}, \quad \forall p \tag{P3-C3}$$

$$\mathrm{tr}(\mathbf{E}_m \mathbf{R}_X) \leq P_{\max}, \quad \forall m \tag{P3-C4}$$

$$\mathbf{W}_k \succeq \mathbf{0}, \quad \forall k \tag{P3-C5}$$

$$\mathbf{Z} \succeq \mathbf{0} \tag{P3-C6}$$

$$\mu_k \geq 0, \quad \forall k \tag{P3-C7}$$

where $\mathbf{A}_k = \frac{1}{\gamma_k}\mathbf{W}_k - \sum_{j \neq k} \mathbf{W}_j$, $\mathbf{F}_p$ is defined in Step 5a, and $\mathbf{E}_m$ is the AP selection matrix.

Table 1: Correspondence between (P1) and (P3) constraints

(P1) constraint	(P3) constraint	Transformation	Convexity
(P1-C1) SINR	(P3-C1) LMI	Steps 1,2,3,4	LMI (convex)
(P1-C2) Sensing SINR	(P3-C2) Linear	Steps 1,2,5b	Linear (convex)
(P1-C3) PCRB	(P3-C3) Linear	Steps 1,2,5a	Linear (convex)
(P1-C4) AP selection	(P3-C4) Fixed set	Step 7	Linear (convex)
(P1-C5) Power	(P3-C4) Linear	Steps 1,6	Linear (convex)
(P1-C6) PSD	(P3-C6) PSD	Unchanged	PSD cone (convex)
(P1-C7) Binary	Eliminated	Step 7	Heuristic
(P1-C8) Rank-one	(P3-C5) PSD	Step 2	PSD cone (convex)

Theorem 1 (Convexity and Equivalence of the Problem)**.** *Problem (P3) is a standard convex SDP. All constraints are LMIs or linear inequalities, and the objective is linear.*

SDR is a lower bound on (P2): $P^*_{P3} \le P^*_{P2} = P^*_{P1}$ (no universal tightness theorem applies—see the Step 1 remark in §4). (P3) can be solved in polynomial time via interior-point methods, with complexity upper bound $O\big((K+1)^3(MN_t)^6 \cdot (K+P+M)\big)$—the precise form follows from the (P3) decision-variable dimension $n = O(K(MN_t)^2)$ and the largest LMI constraint size (MN_t+1); a simplified order is $O((KMN_t^2)^{3.5})$.

Proof. Each constraint in (P3) is either an LMI (P3-C1), a linear inequality (P3-C2, P3-C3, P3-C4), or a PSD cone constraint (P3-C5, P3-C6). The objective is linear. Therefore (P3) is a convex SDP. The transformation chain is:

- **(P1) ⇔ (P2)**: strict equivalence (variable lifting, with the rank-one constraint guaranteeing recoverability).
- **Steps 1–6**: strict equivalences (covariance lifting, non-tight SDR relaxation, S-Procedure equivalence, fractional linearization, PCRB affine rewriting, sensing-SINR linearization, power constraint).
- **Step 2 (SDR)**: a lower-bound relaxation, $P^*_{\mathrm{P3}} \le P^*_{\mathrm{P2}} = P^*_{\mathrm{P1}}$ (no universal tightness theorem applies).
- **Step 7 (AP heuristic)**: the outer AP set is selected by ranking the large-scale fading—this **carries no theoretical optimality guarantee**; it may either yield a lower bound on the original problem or an incomparable objective value if the heuristic picks a suboptimal AP set.

Hence (P3) is a lower bound on (P1): $P^*_{\mathrm{P3}} \le P^*_{\mathrm{P1}}$. The complexity upper bound follows from the standard interior-point result for SDPs (Nesterov-Todd path-following, worst case $O(n^3L)$, where n is the decision-variable dimension and L is the input bit-length). □